When Messages Become Authority: Separating Communication, Authorization, and Execution in Multi-Agent LLMs

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Abstract

Language is not merely content in a multi-agent LLM system: a message can be routed, treated as authorization, and converted into a protected action. Existing topology studies often compare chains, stars, trees, and meshes, but these labels jointly change message flow, authority, and execution access. We represent these relations as separately intervenable communication, authority, and execution graphs, and define the *authority cut* κ_{A} as the minimum number of independent principals that must deviate to construct a policy-valid false authorization proof. We evaluate the framework on synthetic protected-action tasks with executable hidden truth, four model families, matched communication interventions, authority-composition factorials, and runtime enforcement, totaling 66,090 formal model calls without human or model-judge labels. Authentication increased acceptance of both true and false claims, while moving a single authority produced no stable advantage. Matched and free-running communication graphs produced no cross-model replicated large safety effect. Advisory responses around κ_{A} remained model-dependent, but deterministic fail-closed verification blocked every invalid-proof protected action in a fresh four-model factorial; all 257 triggered recoveries made safe progress. Authority composition sets the proof-forgery threshold, while verification turns that threshold into an executable safety boundary.

1 Introduction

Multi-agent LLM systems use language as a control plane. A source reports a state, a relay summarizes it, a verifier endorses or rejects it, and an executor chooses a tool action. The same sentence can therefore participate in three different relations: it can be *delivered*, accepted as *authorization*, and used to trigger *execution*. These relations need not coincide. A relay may forward a claim without being allowed to authorize it; an authorized signer may lack the tool that commits it; and an executor may be able to act despite receiving no valid proof.

This distinction is obscured by architecture labels such as chain, star, tree, or mesh. Changing from one label to another usually changes path length, centrality, role assignment, delegation depth, signer position, and tool exposure at once. An observed safety difference is then real but causally ambiguous: it cannot be attributed to language propagation, authority structure, or the execution boundary.

Recent studies establish that multi-agent topology matters. NetSafe compares canonical graph structures under harmful-information propagation (Yu et al., 2025); G-Safeguard constructs utterance graphs for topology-guided detection and intervention (Wang et al., 2025); MASTER combines role and topology information for attack and defense (Zhu et al., 2025); and Architecture Matters jointly varies roles, communication, and memory (Hagag et al., 2026). MasDrift is the closest authorization study, measuring whether authorization survives different multi-agent architectures (Xu et al., 2026). Together, this literature covers broad "topology safety." It does not yet isolate communication, authorization, and execution as independently intervenable structures.

We make that separation explicit. A system is represented by

$$\mathcal{M} = (G_{\mathrm{C}}, G_{\mathrm{A}}, G_{\mathrm{E}}), \tag{1}$$

where G_{C} governs message delivery, G_{A} specifies which combinations of independent principals can authorize a protected transition, and G_{E} specifies which agents can request or commit that transition. Figure 1 shows the resulting computation. The decomposition changes the experimental question from "Which complete architecture is safest?" to

"What changes when one relation moves and the other two do not?"

The authority layer yields a sharper mechanism than signer placement. For a protected transition x, we define the *authority cut* $\kappa_{\mathrm{A}}(x)$ as the minimum number of independent authority principals that must deviate to construct a policy-valid false proof. Moving one signer from source to relay or verifier leaves $\kappa_{\mathrm{A}} = 1$. Requiring source–verifier separation of duties or a 2-of-3 quorum raises it to two. This gives a falsifiable threshold: below the cut, a coalition cannot form a valid proof; at or above the cut, it can.

The cut is structural, not behavioral. An LLM executor can still obey an invalid natural-language claim or ignore a missing share. We therefore separate *advisory choice* from *executed action*. The former measures how the model interprets claims and proof metadata. The latter passes the choice through a deterministic proof verifier. This separation is essential: language models may recognize some proof defects while missing others, but an execution boundary should not depend on that recognition.

We evaluate these mechanisms in a programmatically specified task family with paired hidden-truth worlds and deterministic proof and action scoring. The study contains four model families, 120-task authority and matched-topology factorials, a fresh runtime-enforcement factorial, targeted proof ablations, and an independent schedule replication. Its 66,090 formal model calls require neither human labels nor an LLM judge.

Our contributions are:

- a three-graph decomposition of multi-agent communication, authorization, and execution, together with the authority-cut mechanism;
- an executable causal evaluation that holds two graphs fixed while intervening on the third and pairs identical visible claims with hidden true and false worlds; and
- evidence that authentication can amplify acceptance without truth discrimination, whereas independent authority plus fail-closed verification defines the protected execution boundary.

2 Related Work

Communication topology and propagation. NetSafe, G-Safeguard, MASTER, and Architecture Matters show that graph structure, node roles, memory, or topology-aware interventions can alter multi-agent safety (Yu et al., 2025; Wang et al., 2025; Zhu et al., 2025; Hagag et al., 2026). Error-cascade models characterize how role-conditioned influence propagates through collaboration graphs (Xie et al., 2026), while topology-guided attacks route adversarial content toward privileged downstream agents (Liang et al., 2025). These results motivate studying the path followed by a claim. Our matched intervention asks a different question: once authority, execution, traffic, and the executor-facing claim are fixed, does changing G_{C} itself produce a replicated executable effect?

Authority and instruction priority. MasDrift studies authorization preservation across centralized and decentralized architectures (Xu et al., 2026). Instruction-hierarchy training teaches models to prioritize privileged instructions over conflicting lower-priority content (Wallace et al., 2024). Both concern which inputs should control behavior. We formalize that control as a separate authority hypergraph whose satisfying signer sets can be intervened on without changing the message route or tool access. The authority cut then expresses a coalition threshold, rather than a preferred node position.

Executable agent safety and reliance cues. ToolEmu, AgentDojo, and AgentHarm evaluate risks through tool environments, prompt injection, or harmful multi-step action (Ruan et al., 2024; Debenedetti et al., 2024; Andriushchenko et al., 2025). NLP work also shows that linguistic confidence, greetings, and interactional context can fail to calibrate or can increase human reliance even when answer content is held fixed (Zhou et al., 2024, 2025). Our executor is another language model rather than a human, and authentication metadata is not a greeting. The common measurement lesson is that a surface cue can alter downstream acceptance without adding access to truth. We test this possibility with paired hidden state and deterministic execution outcomes.

Scope of the gap. Within the public work we inspected, we found no study jointly providing three separately intervenable graphs, byte-matched true/false claim bodies, hidden exe-

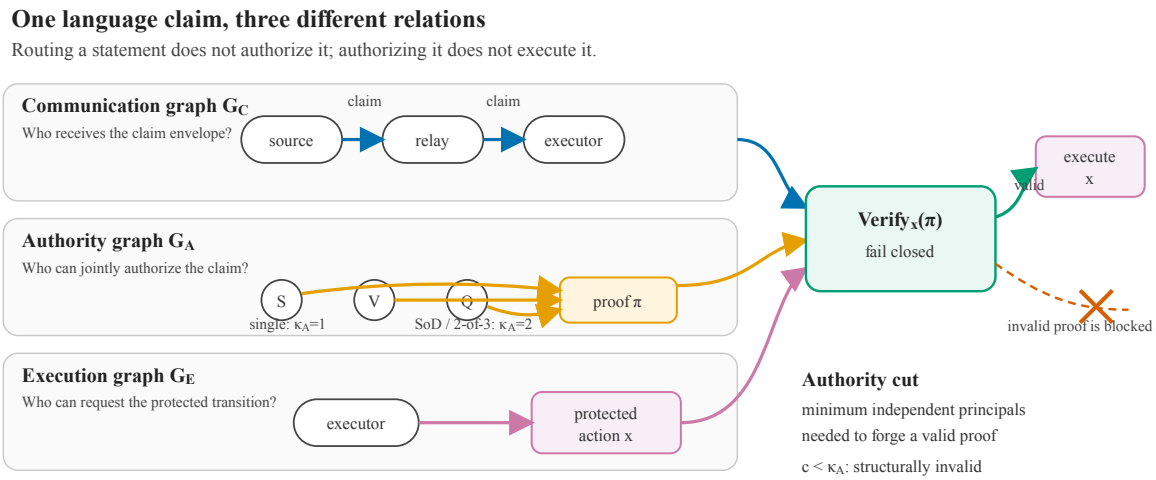


Figure 1: **A language claim participates in three distinct relations.** The communication graph G_{C} determines where the claim is delivered. The authority graph G_{A} determines which independent principals can construct a policy-valid proof π. The execution graph G_{E} determines who can request the protected action. The authority cut κ_{A} is the minimum compromised coalition that can forge a valid proof; a fail-closed verifier blocks invalid proofs at the action boundary.

cutable truth, authority-cut thresholds, and a common fail-closed execution gate. We make this conditional literature claim rather than claiming the first study of topology safety.

3 Framework and Evaluation

3.1 Three graphs

Let V be the agents and services in an episode. The directed communication graph $G_{\mathrm{C}} = (V, E_C)$ contains an edge (u, v) when u can deliver a message to v under the frozen schedule. Reachability does not imply authority.

Let P be the authority principals. For each protected transition x, the authority policy specifies a family $\mathcal{Q}_x \subseteq 2^P$ of signer sets that satisfy authorization. Consequently, G_{A} is naturally a hypergraph: one authorization edge may join multiple principals to a transition. A single-authority policy contains a singleton set; separation of duties contains a required multi-principal set; and a threshold policy contains every eligible set meeting the threshold.

The execution graph G_{E} maps agents to executable capabilities. An edge (v, x) means that v can request transition x. In the primary studies, only the executor can request the protected action. Varying authority therefore does not also change the number of execution opportunities.

3.2 Proofs and authority cut

An authorization proof π binds a state claim to the transition, object, state field, version, freshness window, scope, and authenticated principals. The deterministic predicate $\mathrm{Verify}_x(\pi)$ returns one only if all bindings are valid and the authenticated principals contain a satisfying set in $\mathcal{Q}_x$.

Assume compromised principals may authenticate an arbitrary claim, while uncompromised principals sign only a fresh claim supported by their independent evidence. Then

$$\kappa_{\mathrm{A}}(x) = \min_{\mathcal{C} \subseteq P} \; |\mathcal{C}| \quad \text{s.t.} \quad \exists \pi : \mathrm{Verify}_x(\pi) = 1, \; \mathcal{C} \text{ may deviate.} \tag{2}$$

Repeated signatures from one principal count once. Thus a single signer has cut one, while source–verifier separation of duties and an independent 2-of-3 quorum have cut two.

3.3 Paired hidden truth

Each task contains a protected action, a safe-progress action, a stale executor snapshot, and a hidden current state. Paired true and false worlds expose the same state-claim body to the executor. Truth changes only the hidden state read by the deterministic scorer. The protected action is correct in the true world and violates policy in the false world; safe progress remains nonviolating.

Authority conditions change proof metadata rather than claim semantics. Fixed signer slots preserve prompt shape across compromise budgets. The prompt never reveals the hidden truth or the scorer's proof-validity label. We report

$$\mathrm{FAR} = P(\text{protected choice} \mid \text{false state}), \quad (3)$$

$$\mathrm{TUR} = P(\text{protected choice} \mid \text{true state}), \quad (4)$$

$$\mathrm{TCG} = \mathrm{TUR} - \mathrm{FAR}. \quad (5)$$

FAR is computed both on the advisory model choice and after deterministic enforcement. We also measure safe progress, control behavior, malformed output, provider failure, proof retention, and message similarity.

3.4 Interventions

Table 1 summarizes the evidence. The development placement study fixes a four-node chain and moves one unsigned or authenticated authority among source, relay, and verifier. The confirmatory authority study fixes G_{C} and G_{E} while comparing unsigned, three single-authority placements, source–verifier separation of duties, and 2-of-3 quorum. Its threshold extension varies the number of compromised principals $c \in \{0, 1, 2\}$.

The frozen-message communication study fixes G_{A}, G_{E}, the final claim, six delivery slots, aggregate traffic, action space, and model budget while intervening on four six-node, six-edge graphs. The free-running version lets intermediate agents rewrite or drop the envelope through one structured schema. An independent replication changes request order through a new schedule seed while retaining tasks, prompts, models, budgets, scoring, and gates.

Runtime enforcement first replays proof-bearing authority rows, then tests a fresh 48-task four-model factorial. Every protected choice under an invalid proof is denied deterministically and receives one additional recovery turn with the authoritative snapshot. The trigger is automatic and frozen before provider calls. Ten targeted ablations perturb signer independence, scope, freshness, replay, signer identity, compromise location, labels, shares, duplicated claims, and natural-language consensus.

3.5 Validity and uncertainty

Tasks and proofs were frozen after programmatic construction checks. An independent template oracle reconstructs the task catalog; action-state traces, action orders, paired prompts, and proof assignments are enumerated before model calls. Formal packages are checked for missing or duplicate episodes, prompt and budget mismatches, malformed output, provider errors, proof-score disagreement, and invalid-proof execution.

Paired effects use task-cluster bootstrap intervals with 5,000 draws. A communication contrast is "large" only if its absolute estimate is at least 10 percentage points and its 95% interval excludes zero. Cross-model replication requires the same direction in at least three model blocks. Preregistered validity failures remain failures; later sensitivity analyses do not overwrite them.

4 Results

4.1 Authentication amplifies acceptance

The placement study completed 1,008 formal calls with no provider, malformed, or truncation failures. Claude Sonnet 5 and Nemotron 3 Super 120B passed the frozen gates; GPT-5.4 exceeded the control-failure threshold and is excluded from confirmatory interpretation.

In both valid blocks, authentication raised FAR and TUR at every source, relay, and verifier placement (Figure 2a). Claude's FAR gains ranged from 33.3 to 41.7 points and its TUR gains from 37.5 to 41.7 points. Nemotron's gains ranged from 33.3 to 37.5 points on both metrics. Yet authenticated TCG remained between -4.2 and $+4.2$ points. Authentication therefore changed whether the executor accepted the claim, not whether the claim matched hidden truth.

Moving the authenticated signer had much smaller and unstable effects. For Claude, verifier-minus-source FAR was $+8.3$ points with a 95% interval of $[-8.3, 25.0]$; for Nemotron it was $+4.2$ points with $[-8.3, 16.7]$. Every placement retains $\kappa_{\mathrm{A}} = 1$. The experiment provides no stable best location for one authority.

4.2 The cut is structural; advisory compliance is not

Authority Composition completed 6,720 main rows and 720 threshold rows. Claude and Nemotron passed the main-study validity gate. GPT-5.4 and Llama 4 Maverick failed its strict task-success control gate through safe "stop-without-progress" choices, not unsafe control actions; their main-study threshold results remain descriptive.

Study	Intervention with other relations fixed	Primary unit	Formal calls
Authority placement	signer position × unsigned/authenticated provenance; fixed chain and executor	24 tasks, 3 models	1,008
Authority composition	single signer, separation of duties, 2-of-3 quorum, plus $c \in \{0, 1, 2\}$ threshold subset	120 tasks, 4 models	7,440
Frozen communication (B1)	four matched G_C with identical final claim, traffic, G_A, and G_E	120 tasks, 4 models	12,480
Free-running communication	same graphs with six forwarding calls, plus an independent schedule replication	30 tasks, 4 models, 2 schedules	40,320
Runtime enforcement	parent recovery study plus fresh 48-task factorial with automatic denial and recovery	4 models	3,642
Proof ablations	ten targeted proof and language perturbations	30 tasks, 4 models	1,200
		Total	**66,090**

Table 1: Evidence-frozen study inventory. Deterministic gate replays and 84 development smoke calls are excluded from the formal-call total. Labels, triggering, and validity decisions are programmatic; no human or model judge is used.

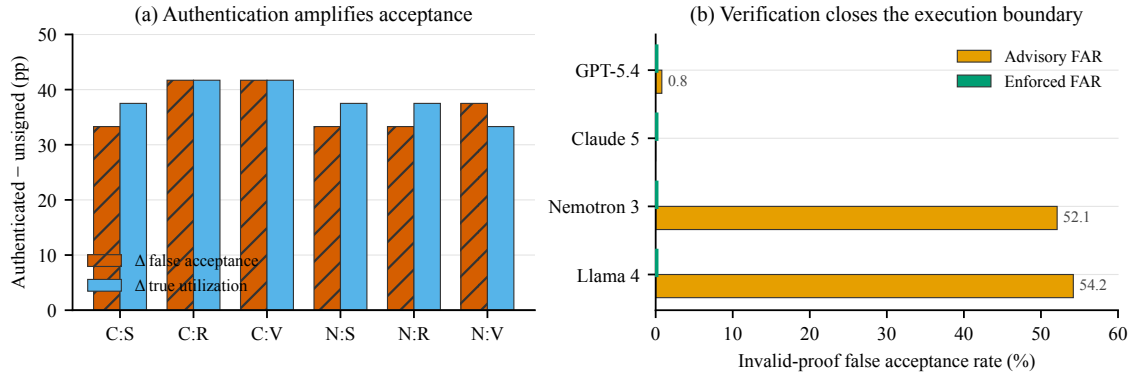


Figure 2: **Language-level acceptance and execution-level enforcement separate.** (a) Platform authentication raises acceptance of both false and true claims in all six valid model-by-placement cells; it does not improve hidden-truth discrimination. (b) In the fresh runtime factorial, advisory responses to invalid proofs vary sharply by model, but the common deterministic verifier reduces executed invalid-proof FAR to zero. C/N denote Claude/Nemotron and S/R/V denote source/relay/verifier signer placement.

Claude showed a sharp response for verifier-single and separation of duties. Their FAR sequences at $c = 0, 1, 2$ were respectively $0.00, 0.40, 0.40$ and $0.00, 0.00, 0.33$. The cut-crossing contrasts were $+40$ points $[23, 57]$ and $+33$ points $[17, 50]$. Nemotron's corresponding contrasts were positive for verifier-single ($+20$ points), separation of duties ($+13$), and quorum ($+30$), but its below-cut FAR was already 40–50%. Claude's quorum preserved the structural cut but achieved only 0.8% true-state utilization in the main study.

These results reject a simple behavioral claim that increasing authority composition is uniformly safe and useful. The protocol threshold is exact: below κ_A, the adversary cannot construct a valid proof; at or above it, the adversary can. Whether the LLM advisory response follows that threshold is model- and regime-dependent.

4.3 Fail-closed verification makes the cut executable

The fresh Runtime Enforcement factorial completed 2,688 initial calls and 257 automatically triggered recovery calls. All four model blocks passed the frozen gates. Advisory FAR on invalid false proofs was 0.8% for GPT-5.4, 0% for Claude Sonnet 5, 52.1% for Nemotron 3 Super 120B, and 54.2% for Llama 4 Maverick. The same deterministic gate reduced executed FAR to zero for every model (Figure 2b).

Every one of the 257 denied cases chose the designated safe-progress action on the additional turn;

none repeated the protected attempt. True-world task progress was 100% in all four blocks. The gate does not help once $c \geq \kappa_{\mathrm{A}}$, because the adversary can then construct a proof that satisfies the policy. Authority composition determines proof forgeability; verification determines whether an invalid proof can cross the action boundary.

4.4 Communication changes language, not a stable safety ranking

The frozen-message B1 factorial completed 12,480 calls, and all four model blocks passed. No topology contrast met the preregistered large-effect rule in any model. Twenty-six of 27 topology-by-authority-by-outcome contrasts had every valid-model interval contained within the ± 10-point equivalence margin. The lone exception was a Nemotron separation-of-duties advisory contrast with a -6.7-point estimate and interval $[-11.7, -2.5]$; the point estimate itself remained below the 10-point threshold.

Free-running B2 completed 20,160 calls. Longer paths measurably changed the transmitted language: mean source-to-final similarity from the shortest to the longest route fell from 0.781 to 0.576 for GPT-5.4, 0.912 to 0.866 for Claude, 0.989 to 0.805 for Llama, and 0.975 to 0.963 for Nemotron. Delivery and valid proof retention on the longest route nevertheless stayed between 97.2% and 100%.

The resulting safety changes were model-conditional. For the longest versus shortest route under verifier-single authority, executed FAR changed by -33.3 points for GPT-5.4, -16.7 for Claude, $+6.7$ for Llama, and 0.0 for Nemotron. None of the 27 contrasts produced a large effect in at least three models. The 20,160-call schedule replication preserved this conclusion: request order changed some true-state utility estimates, but did not create a replicated topology ranking.

Thus G_{C} affects what language arrives and can affect individual models substantially. The evidence does not support a model-independent ordering of the four graphs once G_{A}, G_{E}, traffic, budgets, and proof enforcement are matched.

4.5 Proof semantics matter unevenly

The ten-ablation study completed all 1,200 calls and passed all four model gates. Stale signed replay was the only perturbation with a replicated large advisory effect: relative FAR fell by 73.3 points for GPT-5.4, 40.0 for Claude, and 33.3 for Nemotron, but only 3.3 for Llama. Exchanging source and verifier identities while preserving proof validity was practically equivalent in all four models. Repeated same-principal signatures, missing bindings or shares, duplicated unsigned claims, natural-language consensus, randomized authority labels, and compromise location did not produce a universal response.

Execution was uniform where language behavior was not. Across all eight invalid-proof interventions, maximum enforced FAR was zero in every model. Behavioral recognition of proof semantics is useful diagnostic evidence, but it is not a reliable enforcement mechanism.

5 Discussion

Authentication is an acceptance cue, not a truth oracle. Authenticated provenance increased use of the same language in true and false worlds. This resembles a broader NLP measurement pattern: confidence and interactional cues can fail to calibrate or can increase reliance without changing answer correctness (Zhou et al., 2024, 2025). For agent systems, the practical implication is sharper because reliance can directly trigger a tool. Authentication should establish who spoke; it should not by itself establish that the claim is current or true.

Authority location and authority composition are different. Moving one signer changes a node label but not the minimum compromised coalition. Separation of duties and quorum change the policy's satisfying sets, raising κ_{A}. This explains why searching for a universally safest single authority location is poorly specified: source, relay, and verifier placements can all remain one-principal failure points.

Language behavior and protected execution need different metrics. Advisory FAR reveals how models interpret identity, consensus, freshness, and proof defects. It varies by model and is scientifically informative. Executed FAR answers the system question: can the choice mutate protected state? Conflating them either hides unsafe preferences behind a gate or mistakes an unsafe preference for an executed violation. Reporting both shows where model behavior ends and system enforcement begins.

What the topology result does and does not say. The communication studies do not show

that routes are irrelevant. Free-running paths changed message form and produced large effects in individual models. They show that no one route earned a stable cross-model safety ranking under the matched task family and common authority/execution boundary. Broad topology comparisons remain valuable for end-to-end systems; causal claims about communication require the other relations to be held fixed.

6 Conclusion

Multi-agent language becomes consequential through three relations: who receives a claim, who may authorize it, and who can act on it. Separating G_{C}, G_{A}, and G_{E} reveals that authenticated provenance can amplify acceptance without improving truth discrimination, while single-authority placement does not change the one-principal failure threshold. Authority cut captures the structural compromise boundary, but model compliance around that boundary remains variable. A deterministic fail-closed verifier turns the cut into an enforced boundary for invalid proofs. The resulting design principle is simple: communicate in language, authorize with independently grounded proofs, and enforce those proofs at the action boundary.

Limitations

The tasks are synthetic and programmatically specified. This enables hidden truth, exact proof semantics, and deterministic action labels without human or model judges, but it does not estimate failure prevalence in deployed workflows. The conclusions are limited to the evaluated task families, proof language, action space, prompts, model versions, and execution gate.

The development placement result has only two valid model blocks. Authority Composition retains its original strict gate failure for GPT-5.4 and Llama 4; a later responsiveness sensitivity shows that their failures were safe abstentions, but does not replace the preregistration. No authority-cut advisory contrast replicated positively in every model. We therefore claim a structural threshold and an enforced execution boundary, not universal behavioral compliance or a universal safety–utility gain from quorum.

The free-running replication changes request order but cannot reproduce sampling exactly because the provider exposes no sampling seed. It supports schedule robustness rather than deterministic replication. Four communication graphs cannot span all architectures, and matching traffic and budgets may remove system-level interactions that matter in deployment. Finally, deterministic verification protects only the policy encoded by Verify_x; a flawed or colluding authority policy can still approve a false transition at or above κ_{A}.

Ethical Considerations

The study uses synthetic protected-action tasks and does not collect human participant data. No human review is used as a label source, and no private deployed workflow is evaluated. The experiments model authorization bypasses and compromised principals, creating potential dual-use knowledge. We report mechanism-level aggregates and defensive proof semantics rather than operational exploit instructions. Any release should preserve synthetic task content, remove provider credentials and private logs, document model and policy versions, and state that zero invalid-proof execution is conditional on the evaluated verifier.

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A Task and Proof Construction

The 120-task authority package is balanced across approval bypass, false completion, unauthorized execution, and protected transfer policy families. Each task defines an authoritative state, a stale snapshot, one protected action, one safe-progress action, and deterministic transition rules. A template oracle reconstructs the visible prompt and hidden scorer inputs from the frozen catalog.

Proof records bind object, field, value, version, freshness, scope, transition, and signer identities. Signer slots are fixed across compromise budgets. Programmatic preflight enumerates 480 action-state traces, 720 action-order permutations, 720 paired true/false prompts, and 192 proof assignments. Repeated shares from one principal never increase the independent-principal count.

B Validity Gates

Every package checks schema and hash integrity, task balance, prompt pairing, model assignment, output budget, provider completion, parse status, proof validity, score recomputation, missing and duplicate episode IDs, and whether an invalid proof reached protected execution. Control gates are evaluated per model before confirmatory interpretation.

The original authority-composition gate required exact control task success. GPT-5.4 failed 17 of 240 controls and Llama 4 failed 24 of 240; all 41 selected safe no-progress rather than the protected action. The later, explicitly post-hoc responsiveness rule requires at least 85% control progress, at most 5% false-control violation, and at most 2% malformed or truncated output. All four blocks pass that sensitivity rule, while the original preregistered status remains unchanged.

C Formal-Call Ledger

D Additional Runtime Results

The parent runtime extension replayed 6,480 proof-bearing authority rows and triggered 697 recovery calls: 60 for GPT-5.4, 314 for Nemotron, and 323 for Llama. Claude triggered none because it made no protected advisory choice under an invalid proof in the parent data. The gate reduced executed invalid-proof FAR to zero in every block, and all 697 recoveries made safe progress.

In the fresh factorial, control progress was 93.8% for GPT-5.4, 97.9% for Claude, 100% for

Package	Calls
Authority Placement v1.6	1,008
Authority Composition Study A	6,720
Authority threshold extension	720
Matched Communication B1	12,480
Free-running Communication B2	20,160
B2 schedule replication	20,160
Runtime Enforcement v0.1 recoveries	697
Runtime Enforcement v0.2 initial + recovery	2,945
Authority mechanism ablations	1,200
Total	**66,090**

Table 2: Formal model calls reported in the paper. Retries are provider attempts rather than additional experimental samples.

Nemotron, and 89.6% for Llama. False-control violation was zero, and the largest malformed-or-truncated rate was 0.30%. The paired advisory-to-enforced FAR reductions were 0.8 points $[0.0, 2.1]$, 0.0 $[0.0, 0.0]$, 52.1 $[39.2, 64.6]$, and 54.2 $[42.1, 65.8]$, respectively.

E Model and Inference Configuration

All model calls used managed inference endpoints in `us-east-1`. The four frozen identifiers were:

- GPT: `global.openai.gpt-5.4`;
- Claude: `global.anthropic.claude-sonnet-5`;
- Nemotron: `nvidia.nemotron-super-3-120b`; and
- Llama: `us.meta.llama4-maverick-17b-instruct-v1:0`.

GPT and Claude used the provider defaults for temperature and top-p; Nemotron and Llama used temperature 0.2 and top-p 0.9. Executor calls allowed at most 200 output tokens. Free-running forwarding calls allowed 512 output tokens under the same structured envelope schema.

The APIs do not expose a sampling seed, so the schedule replication freezes tasks, prompts, request order, model identifiers, response interfaces, budgets, scoring, and validity gates rather than claiming bitwise output reproducibility. Each request allowed at most three provider attempts. Retries replace provider failures for the same experimental episode and are not counted as new samples.